

Detecting Intoxication in College Students Using Smartphone Accelerometer Data

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i. Abstract

This project investigates whether smartphone accelerometer data can be used to detect intoxication for the 13 subjects in the *Bar Crawl: Detecting Heavy Drinking* dataset. Transdermal Alcohol Content (TAC) measurements taken from *SCRAM* ankle monitors were used as ground truth. We aligned 120-second windows of smartphone accelerometer readings with the TAC value of the closest timestamp for each participant and created a binary label indicating a subject as sober ($TAC < 0.08$) or intoxicated ($TAC \geq 0.08$). Using this labeled dataset, we trained and evaluated several supervised learning models, namely Logistic Regression, Classification Trees, Random Forests, Bagging, Boosting, Support Vector Machines, and K-Nearest Neighbors, and one unsupervised method, Hierarchical Clustering. Model performance was assessed using accuracy, precision, recall, and F1-score, focusing on how often intoxicated periods were correctly identified. Bagging provided the best overall performance, with 76.94% accuracy, closely followed by Random Forest and Boosting methods. Simpler models such as Logistic Regression and SVM achieved similar accuracies but much lower recall for intoxicated periods, indicating frequent failure to flag episodes of heavy drinking. Hierarchical Clustering and lower-performing supervised models offered limited separation between sober and intoxicated windows. These results suggest that, within this specific bar-crawl event, ensemble tree-based methods can moderately distinguish intoxicated from sober periods using smartphone accelerometer data alone. However, using the small, demographically narrow sample, taken in a single event setting with uncontrolled phone placement, and labeling based on sparse TAC readings mean our findings can only be applied to these 13 participants in this context and are not sufficient to claim that smartphones can reliably detect intoxication in general.

ii. Dataset

We used the *Bar Crawl: Detecting Heavy Drinking* dataset, which we obtained from the UC Irvine Machine Learning Repository (Bar Crawl: Detecting Heavy Drinking [Dataset], 2020). The data was originally gathered by Killian et al. for their 2019 study, *Learning to Detect Heavy Drinking Episodes Using Smartphone Accelerometer Data* (Killian et al., 2019). Their research

focused on evaluating whether smartphones could provide a non-invasive way to identify episodes of heavy drinking in real time.

The study recruited 20 undergraduate seniors (ages 21–23) from The Ohio State University who participated in the university’s annual Bar Crawl on May 2, 2017. Participants were instructed to carry and use their phones “as normal” throughout the event, allowing the researchers to collect ecologically valid data based on naturalistic behavior. Each participant shared their smartphone’s accelerometer data and wore a SCRAM alcohol monitoring ankle bracelet, which recorded Transdermal Alcohol Content (TAC). Due to issues with unusable TAC readings, the final dataset only includes TAC data for 13 of the original 20 subjects.

The accelerometer data was collected at a sampling rate of 40 Hz and is stored in *all_accelerometer_data_pids_13.csv*. Each entry contains a timestamp (in milliseconds), a participant ID (PID), and the accelerometer’s x-, y-, and z-axis measurements. The smartphones used were not standardized: 11 participants used iPhones and 2 used Android devices, as indicated in *phone_types.csv*. Phone placement was uncontrolled, meaning participants could hold, pocket, or bag their phones at any time. This increases ecological validity but introduces additional variability in sensor orientation and movement patterns.

The TAC readings were collected using SCRAM ankle monitors at 30-minute intervals. Raw TAC readings are stored in the *raw_tac* directory, while the *clean_tac* directory contains TAC values that were processed using a zero-phase low-pass filter to reduce noise without causing phase shift and shifted backwards by 45 minutes to align with physiological delays in alcohol excretion through the skin. The cleaned files contain only a timestamp (in seconds) and the filtered TAC value for each reading.

In total, the dataset includes 14,057,567 accelerometer readings and 715 cleaned TAC readings, spread across 13 subjects. TAC and accelerometer files are matched by PID to enable synchronization during analysis.

Several factors limit what can be learned from this dataset. The final sample of 13 participants is small and demographically imbalanced: while the original sample was split evenly by gender, anonymization through PID assignment (combined with the loss of 7 participants) prevents determining whether the final set remains balanced. Seventeen of the 20 original participants identified as white, meaning the dataset likely over-represents this group. Because phone placement and device models vary across subjects, sensor noise and orientation inconsistencies may obscure activity patterns related to intoxication. Additionally, TAC sampling at 30-minute intervals—much slower than the 40 Hz accelerometer data—limits temporal precision when aligning labels.

Despite these limitations, we chose this dataset because it provides the opportunity to study intoxication detection using real smartphone sensor data collected in a naturalistic college drinking environment. As college students ourselves, we considered this dataset particularly compelling as it reflects realistic patterns of behavior in our demographic and provides a meaningful context for exploring predictive models that could support harm-reduction tools for young adults. The combination of high-frequency accelerometer data and validated TAC measurements allows us to investigate whether non-invasive sensor-based detection of heavy drinking is feasible using widely available technology.

iii. Research Question/Problem

In this project, we aim to determine whether we can detect excessive alcohol consumption using only smartphone accelerometer data for the 13 participants included in this study. Heavy drinking poses significant health risks. Long-term effects include liver disease, cardiovascular issues, certain cancers, and mental health disorders (U.S. Department of Health and Human Services, 2025). Short-term consequences including injuries, violence, and alcohol poisoning are similarly dangerous (CDC, 2024). Among college students specifically, approximately 1,519 individuals aged 18–24 die annually from alcohol-related unintentional injuries (U.S. Department of Health and Human Services, 2023). Being able to identify when someone is approaching or experiencing high levels of intoxication offers the potential to intervene earlier and reduce alcohol-related harm.

Our objective is to determine whether accelerometer-based models can identify intoxication ($TAC \geq 0.08$) without relying on direct biochemical measurement. TAC is commonly used as a proxy for Blood Alcohol Content (BAC), and 0.08 corresponds to the legal limit for operating a motor vehicle in the United States (*Adult Operators of Noncommercial Motor Vehicles*, 2024). By examining whether movement patterns recorded through smartphone sensors correlate with elevated TAC levels, we hope to evaluate the feasibility of a non-invasive, automated detection system.

Scholars have previously examined this question. Killian et al. (2019) analyzed this dataset using Multi-Layer Perceptron (MLP), Convolutional Neural Network (CNN), Support Vector Machine (SVM), and Random Forest (RF) models (Killian et al., 2019). Their analysis used low-pass filtering to smooth TAC and accelerometer signals and applied a binary classification threshold of $TAC < 0.08$ (sober) vs. $TAC \geq 0.08$ (intoxicated). Among the models tested, Random Forest performed best with an accuracy of 77.48%.

Beyond this study, researchers have investigated alternative non-invasive intoxication detection methods. For example, a team at Iowa State University developed an infrared deep learning system using facial heat signatures to classify intoxication status and achieved an accuracy of 74.07% in binary classification tasks (Acharya & Radhakrishnan, 2024). Although their approach uses image-based rather than accelerometer data, it demonstrates broader scholarly interest in automated intoxication detection and highlights the need for more accessible, widely deployable systems.

Our project builds directly on the work of Killian et al. We focus on implementing and evaluating three supervised learning methods that were not tested in the original study: Logistic Regression (LR), Classification Trees, and K-Nearest Neighbors (KNN). We also propose analyzing Bagging and Boosting methods, which extend tree-based models through ensemble techniques. Additionally, we plan to apply Hierarchical Clustering (HC) as an unsupervised approach to explore whether sober and intoxicated periods exhibit separable structure when labels are ignored. Our goal is to determine whether these new methods provide improved performance and observe how our results compare to those reported in prior research.

iv. Research Methods

This dataset directly supports our research question because it contains synchronized smartphone accelerometer readings and TAC measurements collected during real episodes of drinking. The TAC values provide labels for intoxication, while the accelerometer readings capture movement patterns that may correlate with changes in motor control due to alcohol consumption. By linking TAC readings to corresponding accelerometer windows, we can train both supervised and unsupervised machine learning models to determine whether intoxication can be inferred from motion data alone.

Data Preprocessing

Before analysis, the dataset required multiple steps of preprocessing. TAC readings were stored across 13 separate CSV files, one per subject, and were consolidated into a single dataset containing PID, timestamp (in seconds), and cleaned TAC values. We converted the TAC readings into a binary classification variable where readings < 0.08 were labeled 0 (sober), and readings ≥ 0.08 were labeled 1 (intoxicated). This threshold reflects the legal BAC limit in the United States and follows the classification standard used in the original study (*Adult Operators of Noncommercial Motor Vehicles*, 2024; Killian et al., 2019).

The accelerometer file contains over 14 million readings sampled at 40 Hz, which is too large to load directly into R. Thus, we had split the file into 15 separate CSV files, each containing approximately one million rows. For each row, we computed the magnitude of the acceleration vector using the following formula:

$$m = \sqrt{x^2 + y^2 + z^2}$$

To align with the TAC sampling rate and reduce file size, we aggregated readings into 120-second windows for each PID. Although the original paper tested multiple window sizes and found that 10-second windows yielded the highest accuracy, it also reported that 60-second and 120-second windows performed comparably well. Given our constraints with memory, we selected 120-second intervals as a balance between accuracy and feasibility.

After aggregating across all 15 accelerometer files—using weighted averaging for windows spanning across split CSV files—we converted the accelerometer timestamps from milliseconds to seconds to align them with the TAC timestamps. Each 120-second window was assigned a binary sobriety label based on the TAC reading closest in time for that PID. Missing TAC values were removed from the processed dataset, and any accelerometer windows lacking a corresponding TAC reading were labeled using the temporally closest TAC measurement.

Proposed Methods

We propose using a combination of supervised and unsupervised machine learning techniques. These techniques are well-suited for our data and our research objectives for the following reasons:

Supervised Methods

Logistic Regression:

LR is an interpretable baseline model for binary classification, appropriate for determining whether linear combinations of accelerometer features predict intoxication. It provides coefficients that help identify which aspects of movement are most associated with elevated TAC.

Classification Trees:

Trees capture nonlinear relationships and interactions between features. This is important because intoxication may affect movement patterns in complex ways not captured by linear models. Trees also handle mixed distributions and are robust to outliers.

Random Forest and Bagging:

These ensemble methods reduce variance by averaging across many trees, improving generalization. Bagging allows us to explore whether aggregating multiple trees yields more stable intoxication predictions compared to a single tree.

Boosting:

Boosting fits trees sequentially, allowing the model to focus on harder-to-classify cases. This is especially relevant given the imbalance between high-frequency accelerometer data and infrequent TAC sampling. We plan to tune the number of trees, interaction depth (1–4), and cross-validation folds (2–7), selecting parameters that maximize accuracy without overfitting.

K-Nearest Neighbors:

KNN is a non-parametric method that classifies based on similarity in feature space, making it suitable for detecting subtle shifts in movement patterns. We plan to test odd k-values from 3 to 17 to avoid ties and capture a range of neighborhood sizes. Smaller k values may overfit noise, while larger ones may oversmooth distinctions between sober and intoxicated movement.

Support Vector Machines:

SVMs identify the decision boundary that maximizes the margin between sober and intoxicated observations, often leading to strong generalization on high-dimensional data. By using kernel functions, SVMs can capture nonlinear relationships in accelerometer features that may distinguish subtle differences in movement caused by intoxication.

Unsupervised Methods

Hierarchical Clustering:

HC allows us to explore whether sober vs. intoxicated segments naturally form separable clusters without labels. Because HC is sensitive to distance metrics and linkage methods, we will evaluate multiple linkage strategies (e.g., complete, average, single) to assess cluster structure.

Parameter Tuning

Many models include hyperparameters that can significantly influence performance:

Bagging:

We will test mtry values from 1–4, reflecting the number of predictors available.

Boosting:

We will use cross-validation to tune the number of trees, tree depth, and cv folds.

SVM:

We will compare kernel types to evaluate how decision boundary shape affects performance.

KNN:

We will test odd k values from 3–17 to balance bias and variance.

HC:

We will compare linkage methods to assess cluster separability.

After implementing these models, we will compare accuracy, precision, recall, and F1-score across all methods, and evaluate how our results align with or differ from those obtained by Killian et al.

Table

The table below indicates which of these proposed plans we followed through on in this project, ultimately leading to a ‘story.’

Task	Completed
LR Model	✓
Tree Model	✓
RF Model	✓
Bagging Model	✓
Boosting Model	✓

KNN Model	✓
SVM Model	✓
HC Model	✓
Bagging Tuning	✓
Boosting Tuning	✓
SVM Tuning	✓
KNN Tuning	✓
HC Tuning	✓

v. Project plan

To organize our work, we will divide tasks evenly and collaborate through a shared R Markdown document, which will serve as the basis for our final report. We plan to meet twice weekly to follow our planned timeline and ensure we remain on track. Each meeting will focus on reviewing work completed since the previous meeting, comparing notes, and outlining concrete goals to accomplish before the next session. Throughout the project, we will rotate responsibilities such as scheduling meetings, documenting progress, locating academic sources, and drafting deliverables. This rotation ensures that each team member contributes to every component of the project while allowing us to adapt responsibilities depending on the phase of the project.

To complete this project effectively, we will draw on a variety of statistical techniques we have learned in class, such as logistic regression, decision trees, and k-nearest neighbors. In addition, because our dataset requires substantial manipulation before it is usable, we will need to develop new skills in data processing and cleaning, particularly working with time-series sensor data, aligning sampling rates across devices, and preparing condensed windows for modeling.

Timeline of Tasks

Monday 11/17

Finalize research methods and statistical techniques

Draft the Research Methods section

Complete necessary data manipulation and preprocessing

Friday 11/21

Check assumptions for selected statistical techniques

Run analyses with chosen models

Generate visualizations and summary statistics

Monday 11/24

Write the “Analyze Data / Results” section of the report

Friday 11/28

Thanksgiving break, no meeting

Monday 12/1

Write the Summary and Abstract sections

Complete report formatting and integration of all sections

Friday 12/5

Create and finalize the class presentation

Monday 12/8

Present in class

Friday 12/12

Submit Final Report

vi. Analyze Data and Results

Model Performance and Initial Interpretation

The models we chose are appropriate because they allow us to test whether intoxication affects movement in linearly separable ways, distance-based ways, or through more complex nonlinear patterns.

Table 1 summarizes the classification performance of all models on the held-out test data in terms of accuracy, precision, recall, and F1-score. Bagging achieved the best overall

performance, with an accuracy of 0.7694, precision of 0.6169, recall of 0.6966, and F1 of 0.6544. Random Forest and Boosting performed almost identically, with accuracies of 0.7676 and 0.7606 and F1-scores of 0.6489 and 0.6304, respectively. Logistic Regression, Support Vector Machines, and K-Nearest Neighbors showed moderate accuracies (0.69–0.71) but substantially worse F1-scores, primarily due to low recall for intoxicated periods. Hierarchical Clustering performed worst across all metrics.

These results provide an initial answer to our research question. Tree-based ensemble methods appear capable of detecting intoxication from short accelerometer windows with moderate accuracy, while linear and distance-based models struggle to reliably pick up intoxication-related movement changes.

Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.6954	0.5472	0.1629	0.2511
Classification Tree	0.6567	0.4719	0.8034	0.5946
Random Forest	0.7676	0.6162	0.6854	0.6489
Bagging	0.7694	0.6169	0.6966	0.6544
Boosting	0.7606	0.6105	0.6517	0.6304
Support Vector Machine	0.6972	0.5625	0.1517	0.2389
K-Nearest Neighbors	0.7130	0.5510	0.4551	0.4985
Hierarchical Clustering	0.6331	0.3459	0.0719	0.1191

Table 1: Test-set performance metrics for all models

Figure 1 displays these results as a bar chart of accuracy by model. This visualization makes the ranking clear and is appropriate here because it allows for straightforward comparison across models. Bagging, Random Forest, and Boosting form the top-performing group, followed by Logistic Regression, SVM, and KNN, and the single Classification Tree and Hierarchical Clustering lag behind. From this visualization alone, tree-based ensemble methods appear to be the most effective for this dataset. This ranking reinforces the idea that intoxication-related movement changes are nonlinear and subtle, since models that can capture interactions and irregular patterns outperform simpler linear or distance-based models. This suggests that

intoxication does create detectable yet complex signatures in accelerometer data, consistent with our research question.

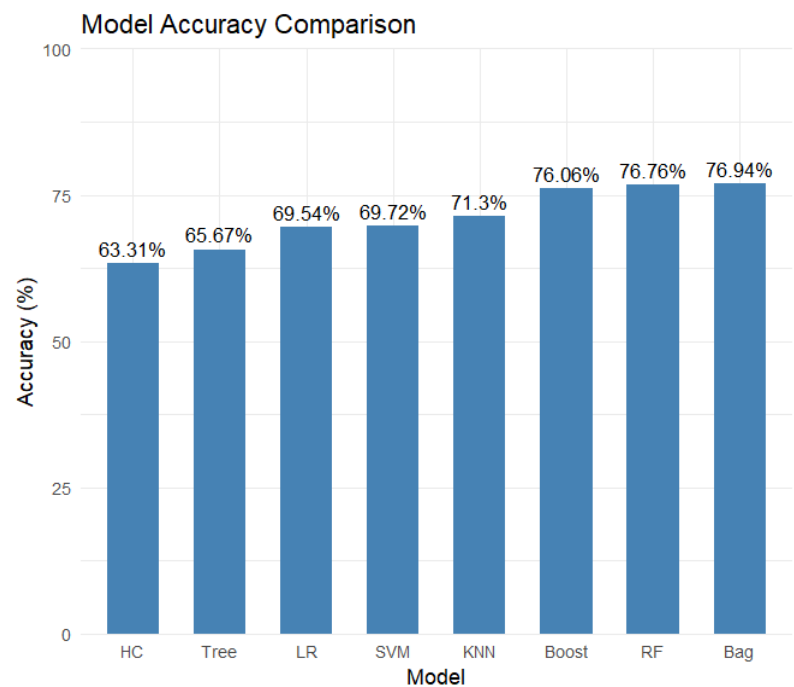


Figure 1. Test accuracy (%) for each model

This visualization reinforces the pattern from Table 1 by clearly showing that the models best able to capture complex relationships (Bagging, RF, Boosting) also produce the highest performance.

Error Patterns and Real-World Meaning

Figure 2 shows confusion matrices for Logistic Regression, Bagging, and the Classification Tree. All models share the same underlying class distribution (390 sober windows vs. 178 intoxicated windows), so differences reflect model behavior rather than data imbalance. Presenting confusion matrices is valuable because they highlight not only overall accuracy but also the specific types of errors each model tends to make.

Logistic Regression			Classification Tree			Bagging		
Accuracy: 0.6954			Accuracy: 0.6567			Accuracy: 0.7694		
Precision: 0.5472			Precision: 0.4719			Precision: 0.6169		
Recall: 0.1629			Recall: 0.8034			Recall: 0.6966		
F1: 0.2511			F1: 0.5946			F1: 0.6544		
	True			True			True	
Predicted	sober	intoxicated	Predicted	sober	intoxicated	Predicted	sober	intoxicated
sober	366	149	sober	230	35	sober	313	54
intoxicated	24	29	intoxicated	160	143	intoxicated	77	124

Figure 2: Confusion matrices and test metrics for Logistic Regression, Classification Tree, and Bagging

Logistic Regression attains reasonable accuracy (0.6954) but very low recall for intoxicated windows (0.1629). Out of 178 intoxicated windows, it correctly identifies only 29 and misclassifies 149 as sober, meaning a logistic model would miss most intoxicated episodes. In the context of our research question, this means a logistic model would miss nearly half of all real intoxication episodes, making it unsuitable for intoxication detection in naturalistic environments.

The Classification Tree behaves oppositely, with lower accuracy (0.6567) and low precision (0.4719) but high recall (0.8034). It identifies most intoxicated periods but mislabels many sober windows as intoxicated. This would produce many false alarms in real-world use, potentially undermining trust in a detection system.

Bagging strikes a middle ground, correctly identifying 124 of 178 intoxicated windows (recall 0.6966) while maintaining higher precision than the single tree. Its F1-score (0.6544) is the highest among all models. From a practical standpoint, Bagging provides the most balanced trade-off between missed intoxicated periods and unnecessary intoxication alerts. However, its error rates are still too high for deployment in any safety-critical context. Thus, for our dataset, accelerometer data does contain useful intoxication-related information, but the signal is too inconsistent for high-stakes decision-making.

Together, the confusion matrices and metrics show that simple linear models under-detect intoxication, single trees over-flag intoxication, and ensemble trees yield the most reasonable compromise for this specific dataset.

Variable Importance and Model Behavior

Figure 3 reports variable-importance measures for Bagging and Random Forest. Across both models, the magnitude of acceleration m is consistently the most important predictor, with x- and

z-axis components providing secondary contributions and y-axis contributing least. This pattern is consistent for both MeanDecreaseAccuracy and MeanDecreaseGini in the tree-based models, and for relative influence in the Boosting model. These plots suggest that overall movement intensity, rather than any single directional component, carries the strongest signal about intoxication in these 120-second windows.

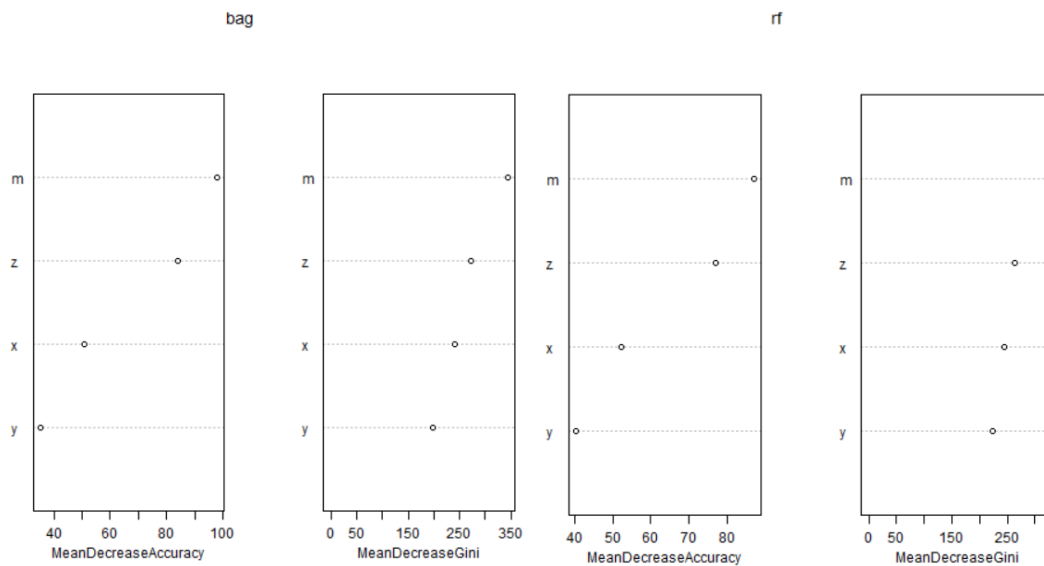


Figure 3: Figure 3. Variable importance (MeanDecreaseAccuracy and MeanDecreaseGini) for Bagging (left) and Random Forest (right)

These importance plots visually support the earlier classification results by illustrating that overall movement intensity, rather than any single directional component, carries the strongest signal about intoxication in these 120-second windows. This aligns with intuition: as people become more intoxicated, their stability and gait consistency break down, producing irregular magnitude fluctuations. The data and models therefore support the idea that accelerometer magnitude is a meaningful digital biomarker of intoxication, which strengthens the biological plausibility of our results.

Tuning Diagnostics and Assumptions

Figure 4 shows the cross-validated error for Boosting as a function of the number of trees. The error decreases rapidly at first and then flattens, with the minimum around 116 trees, which we adopt as the final model size.

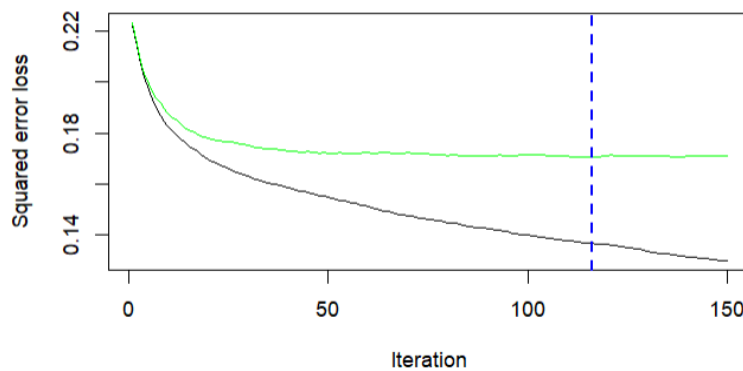


Figure 4. Cross-validated squared error loss for Boosting as the number of trees increases (vertical line indicates the selected 116 trees)

Table 2 summarizes tuning results for Bagging (mtry), Boosting (interaction depth and cv.folds), SVM (kernel choice), KNN (k), and HC (linkage method). For Bagging, accuracy peaks at mtry = 3, supporting the choice used in the final model. For KNN, accuracy improves steadily with k and plateaus around k = 15. For SVM, the sigmoid kernel slightly outperforms polynomial and radial kernels, but all three achieve similar and relatively modest accuracy. For HC, different linkage methods (single, average, complete) all perform poorly, confirming that sober and intoxicated windows do not form well-separated clusters in this feature space.

Model	Hyperparameter(s)	Best Value(s)	Best Accuracy
Bagging	mtry	3	0.7694
Boosting	n.trees, interaction.depth, cv.folds	116 trees, depth = 4, folds = 3	0.7606
SVM	kernel	sigmoid	0.6972
KNN	k	15	0.7130
Hierarchical Cluster	linkage method	single/avg/complete (similar)	0.6331

Table 2. Hyperparameter tuning results for Bagging, Boosting, SVM, KNN, and Hierarchical Clustering

These tuning plots and tables justify the selected hyperparameters and show that performance differences across models are not artifacts of arbitrary parameter choices.

Regarding assumptions, logistic regression assumes a linear relationship between predictors and the log-odds of intoxication and independent observations. Neither assumption strictly holds here: repeated windows from the same individual introduce dependence over time, and the relationship between movement patterns and intoxication is likely nonlinear. The weak recall of the logistic model is consistent with these violated assumptions. Tree-based methods, Bagging, Random Forests, and Boosting make fewer distributional assumptions and handle nonlinear interactions and complex decision boundaries, which explains their superior performance. KNN and SVM operate directly in feature space and rely on a meaningful distance metric; because all predictors are measured in the same physical units (components and magnitude of acceleration), Euclidean distance is reasonable, but the limited number of features restricts how well these methods can separate classes. These assumption checks support the idea that tree-based ensemble models are better suited to realistic, messy smartphone movement data, while linear and distance-based models are too restrictive.

Finally, all evaluation metrics are computed on a test set drawn from the same 13 subjects used for training. This means the reported performance reflects “how well we can detect intoxication for these 13 people during this specific bar crawl,” not intoxication detection in the general population. The visualizations and numerical results reinforce this interpretation: ensemble tree models can moderately distinguish sober from intoxicated windows in this dataset, but error rates and sampling limitations prevent strong claims about general smartphone-based intoxication detection. Additionally, because our train/test split was performed at the window level rather than by subject, windows from the same participant may appear in both sets. This can cause models to perform better than they would in a fully subject-independent setting, and we treat this as an important limitation.

vii. Summary

The goal of this project was to determine whether smartphone accelerometer data could identify intoxication in real time for the 13 participants of the Bar Crawl dataset. By pairing 120-second windows of accelerometer magnitude with TAC-derived sobriety labels, we

evaluated a broad set of supervised and unsupervised learning models to assess the extent to which movement patterns reflect intoxication levels ($TAC \geq 0.08$). Our results provide moderate evidence that such detection is possible, but also illustrate substantial limitations that restrict generalizability beyond this specific event and group of participants.

Our analysis required extensive preprocessing, including consolidating TAC files, computing acceleration magnitudes, splitting the accelerometer dataset for memory efficiency, and aggregating raw sensor readings into windowed summaries. After aligning accelerometer windows with the closest TAC timestamps, we trained Logistic Regression, Classification Trees, Random Forests, Bagging, Boosting, SVMs, and KNN models, along with Hierarchical Clustering as an unsupervised baseline. We tuned hyperparameters to ensure fair comparison and evaluated performance using accuracy, precision, recall, and F1-score, with particular emphasis on the correct identification of intoxicated periods.

Across all supervised models, tree-based ensemble methods consistently performed best. Bagging achieved the highest accuracy (76.94%) and F1-score (0.6544), followed closely by Random Forest and Boosting. These models were also able to identify a greater proportion of intoxicated windows compared to linear or distance-based methods. Simpler models such as Logistic Regression and SVM showed adequate accuracy but extremely low recall, meaning they frequently failed to detect intoxication. KNN performed slightly better but still lagged behind ensemble trees, while Hierarchical Clustering showed minimal separation between sober and intoxicated periods, indicating that intoxication-related movement patterns are not strongly distinguishable in an unsupervised setting.

The variable-importance results further illuminate model behavior: across Bagging, Random Forest, and Boosting, acceleration magnitude was consistently the most influential predictor. This aligns with biological expectations—alcohol impairs coordination, producing less stable and more erratic movement patterns. Thus, the data support the presence of a meaningful relationship between intoxication and overall movement intensity.

Despite these promising results, several factors restrict what we can conclude. The dataset is small, demographically narrow, and collected in a single highly specific environment—a college bar crawl. Phone placement was uncontrolled, TAC readings were sparse relative to accelerometer sampling, and repeated windows within individuals introduced dependence that most models do not fully account for. As a result, our findings answer our original research question only in a limited sense: for these 13 participants, during this one event, ensemble tree-based models can moderately detect intoxication using smartphone accelerometer data.

The broader claim that smartphones can reliably detect intoxication in general populations cannot be supported. Instead, our results illustrate both the potential and the substantial challenges of accelerometer-based intoxication detection. Future work should incorporate larger and more diverse populations, standardized phone placement, additional sensor modalities, and more advanced temporal modeling techniques to move toward practical, real-world applications.

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